

Analysis of Greenhouse Gas Emissions

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STT 461

Introduction/Why

- We analyze fossil fuel use, production, and leakage
- Compare different plant types
- Understanding emission sources is critical
- Identifying where emissions come from helps guide reducing efforts

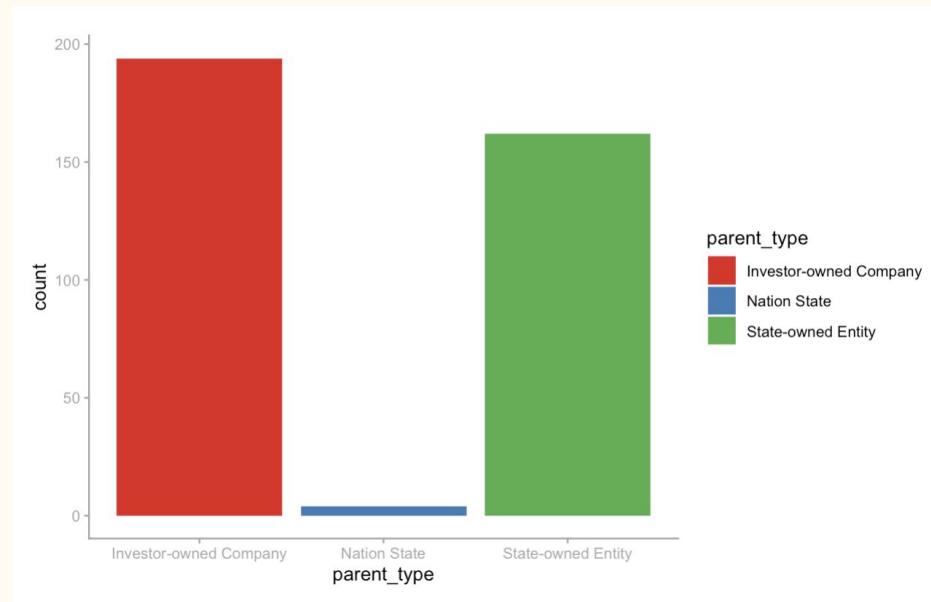
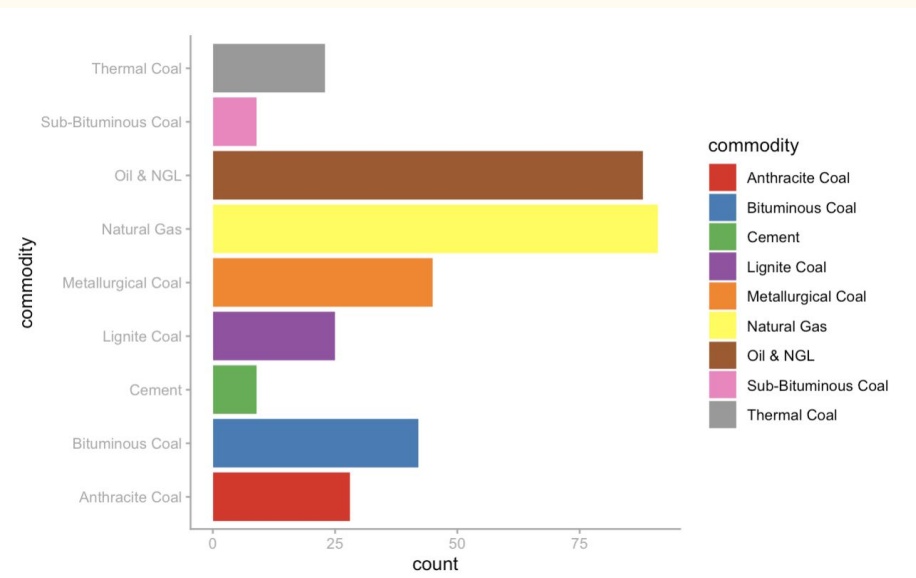


Data Information

- Source: Carbon Majors Dataset (carbonmajors.org)
- 2024 data only
- Selected variables:
 - Commodity
 - Parent type
 - Total emissions
 - Fugitive emissions
 - Own fuel use
- Final dataset: 360×6



Exploratory Analysis



Methodology

- **Bootstrap Hypothesis Test**
 - Resample data
 - Simulate null distribution
 - Compute p-value
- **Bootstrap Confidence Intervals**
 - Resample data
 - Build distribution
 - Take percentiles
- **Kernel Density Classifier**
 - 80% training, 20% testing
 - Naive Bayes Classifier with 5-fold cross-validation
 - Model for CO₂ emissions and fugitive emission



Research Questions

1,2,3 -Mean total emissions of Natural Gas plants and Oil & NGL

4,5 -Mean fugitive emissions of Natural Gas plants and Oil & NGL

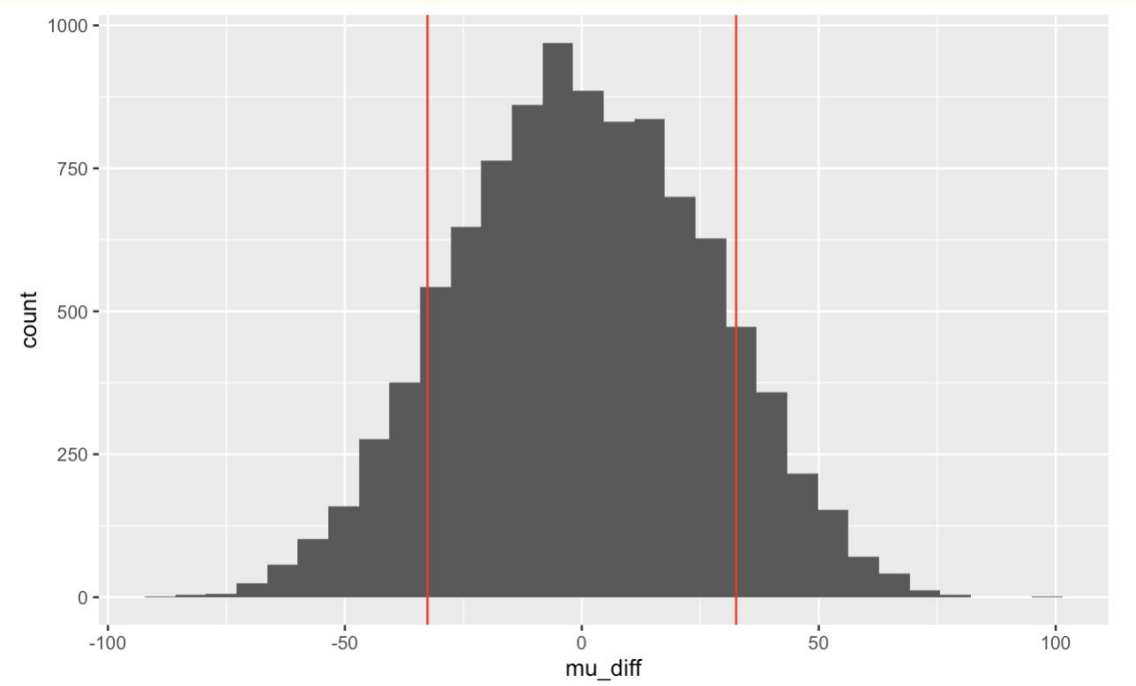
6 -Mean own fuel use emissions of state owned and investor owned

7 -Naive Bayes Classification

Mean total emissions of Natural Gas plants and Oil & NGL

1. Are the mean total emissions of Natural Gas plants significantly different from Oil & NGL plants?
2. What is the 95% confidence interval for the mean total emissions of Natural Gas plants?
3. What is the 95% confidence interval for the mean total emissions of Oil & NGL plants?

1. No Significant Difference in Total Mean Emissions



P Value=.2286

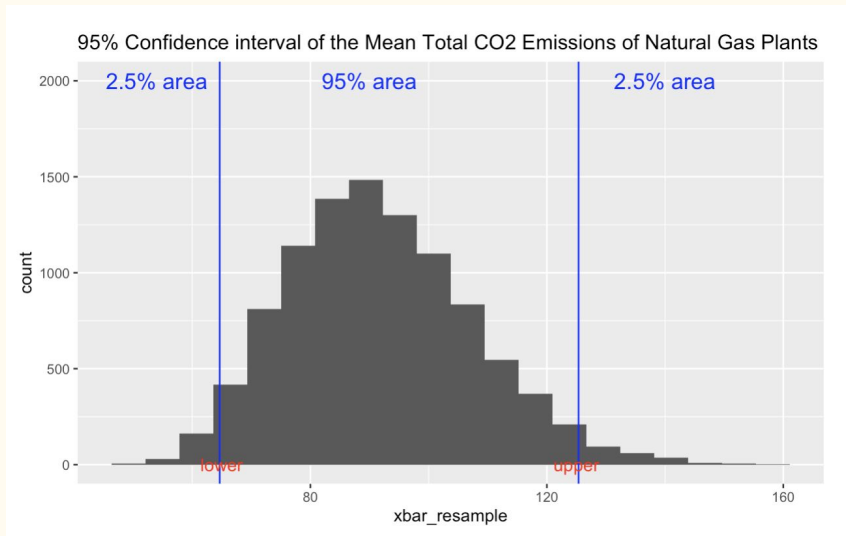
Not Statistically Significant

Not enough evidence to
conclude there is difference

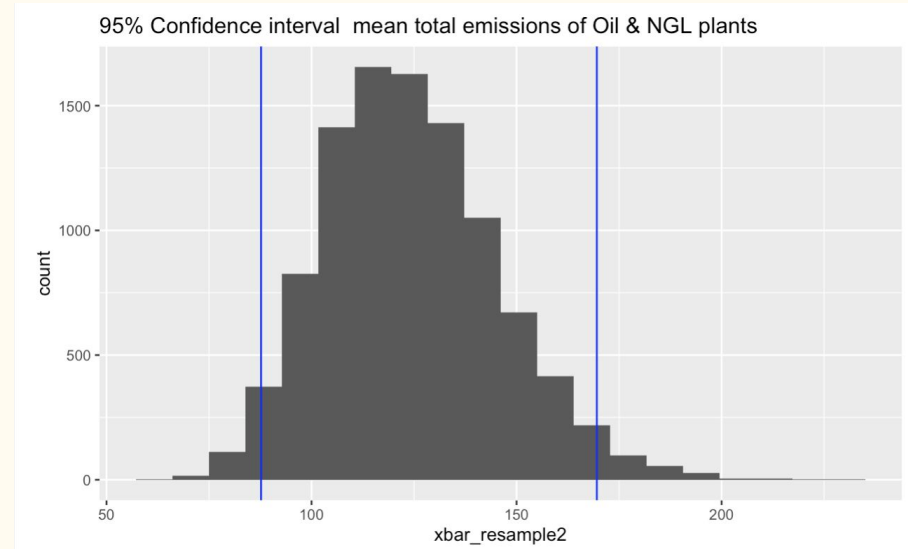
2. 95% confidence interval for the mean total emissions of Natural Gas plants?

3. 95% confidence interval for the mean total emissions of Oil & NGL plants?

2. (64.65603,125.36208) million metric tons.



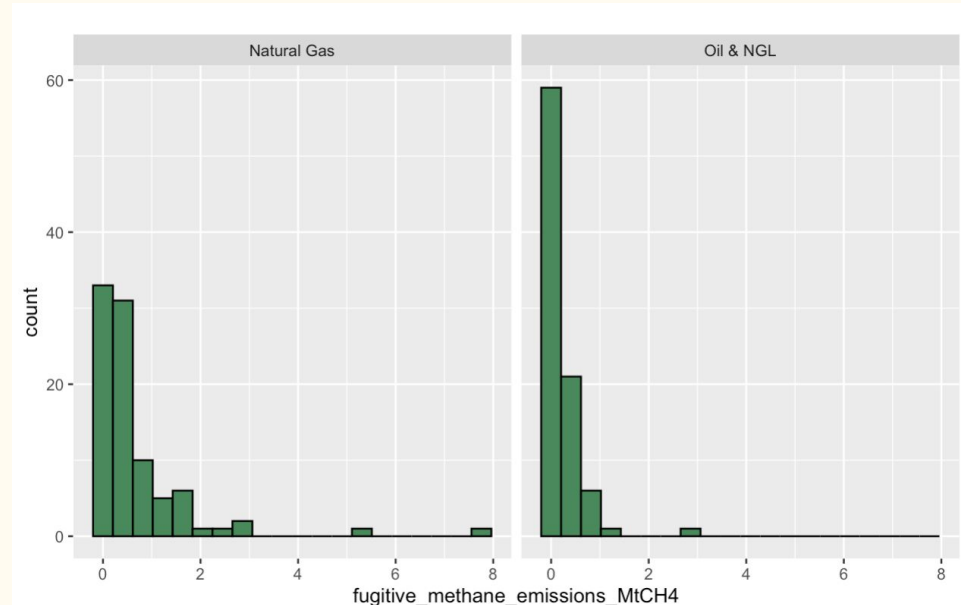
3. (87.69647 ,169.57829) million metric tons



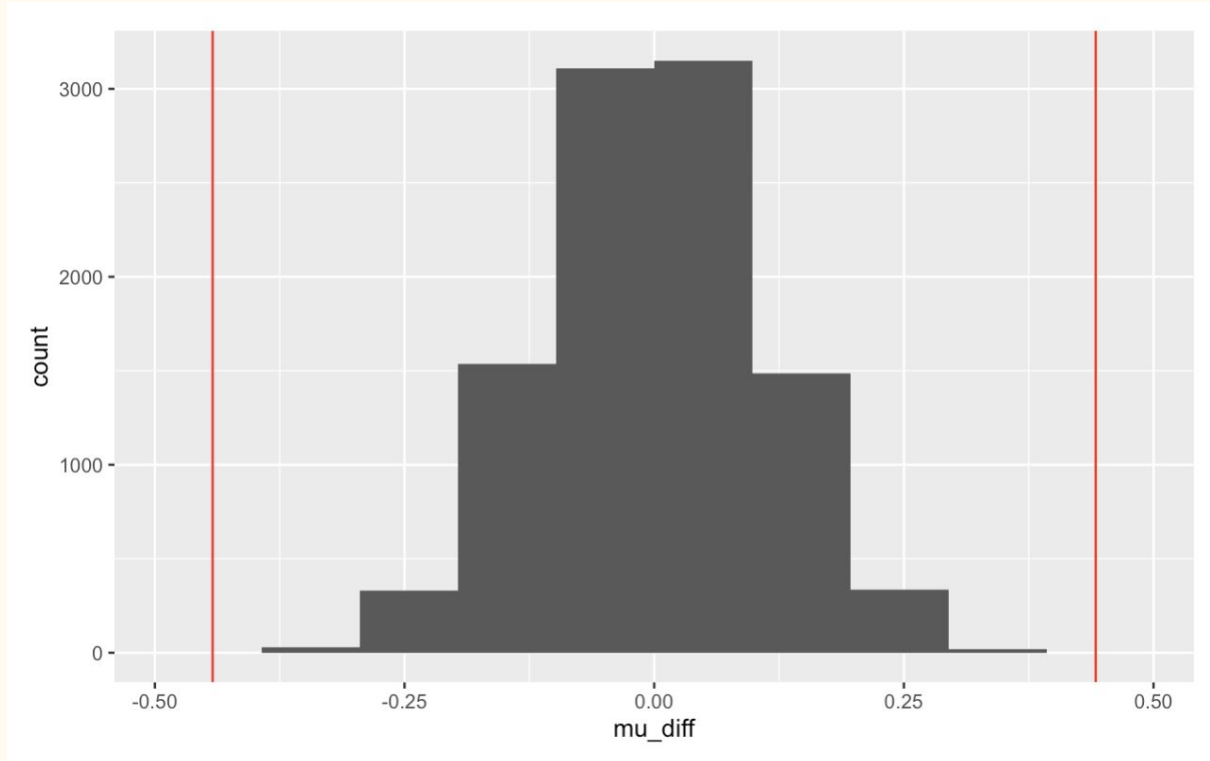
Mean fugitive emissions of Natural Gas plants and Oil & NGL

4. Are the mean fugitive methane emissions of Natural Gas plants significantly different from Oil & NGL plants?

5. 95% confidence interval for the difference of mean fugitive methane between Natural Gas Plants and Oil & NGL



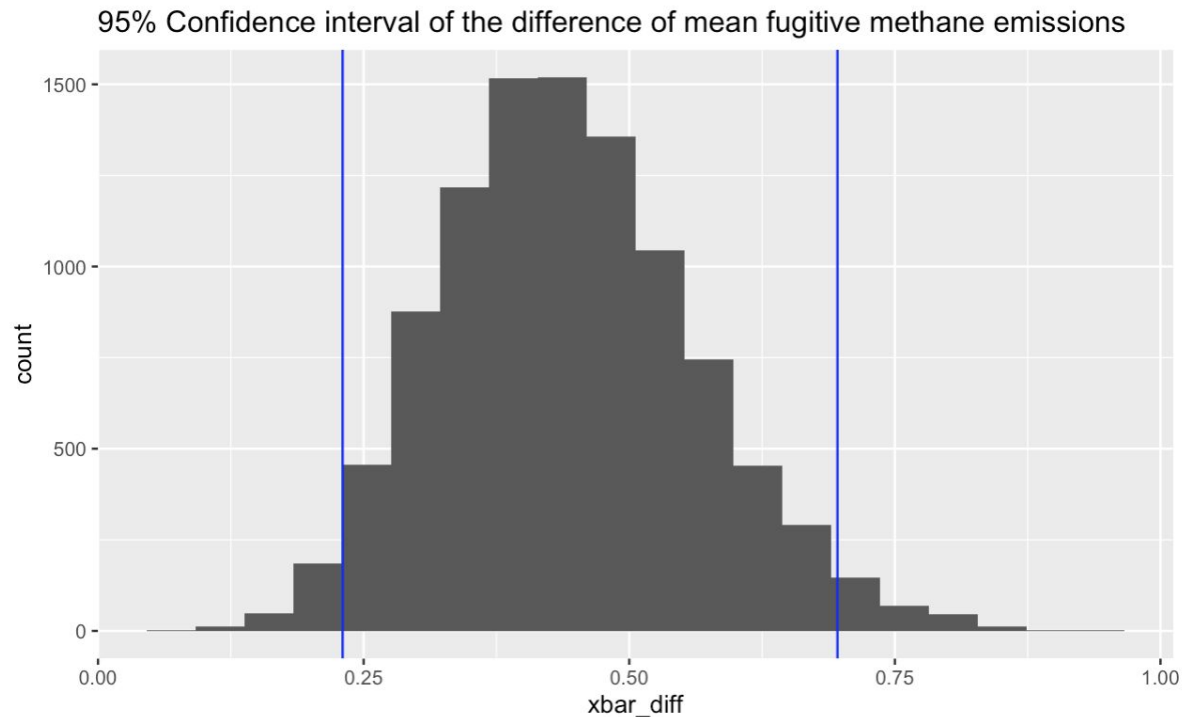
4. Significant Difference in Mean Fugitive Emissions



P-Value of <0.00001

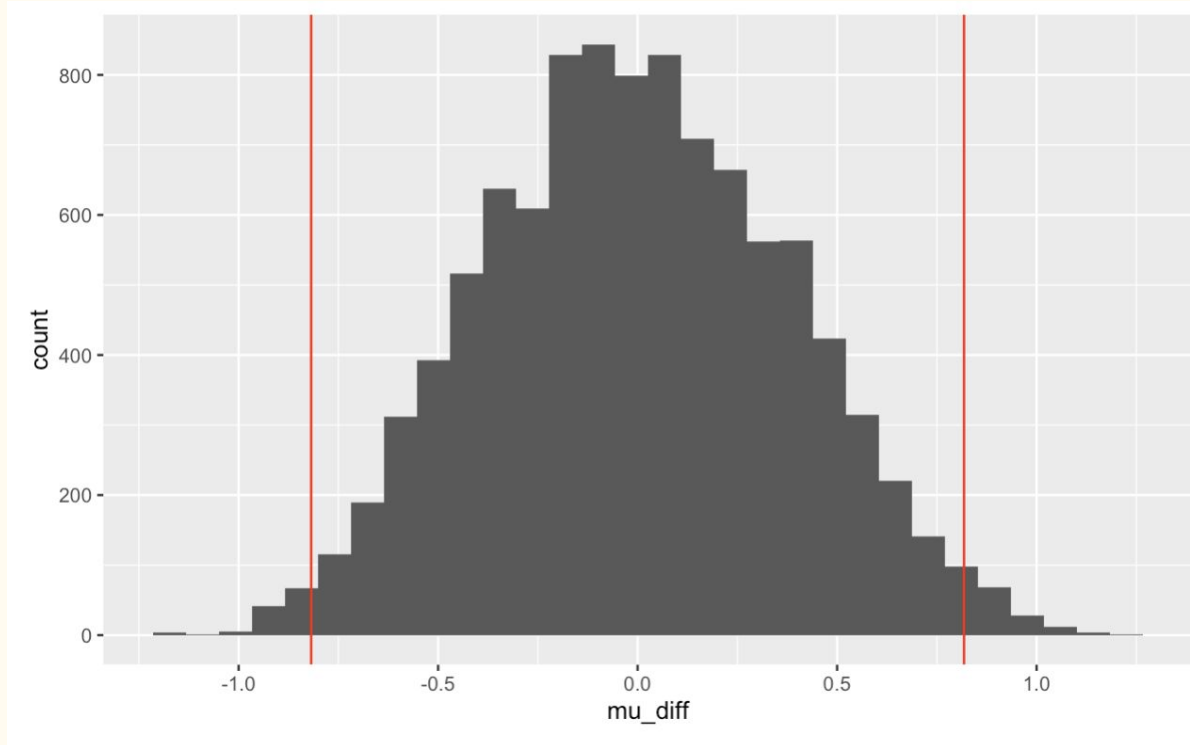
Very strong evidence of difference

5. 95% confidence interval for the difference of mean fugitive methane between Natural Gas Plants and Oil & NGL



(0.2301867 ,0.6960284)
million metric tons.

6. Difference in mean own fuel use emissions



P-Value of 0.0244

**Strong Evidence of
Difference between State
Owned and Investor
Owned**

7. Given an entity's total CO₂ emissions or fugitive methane emissions, can you predict whether it came from a Natural Gas or Oil & NGL plant?

- The CO₂ model kappa= **0.032**, **poor agreement**
- The fugitive methane model kappa=**0.207**, **fair agreement** between the true and predicted values,
 - slight predictive power
- **Fugitive methane is a more reliable predictor** than CO₂ when classifying between Natural Gas and Oil & NGL plants

Conclusions

1. No significant difference between CO₂ emissions between Natural Gas and Oil & NGL
2. Evidence of difference in fugitive emissions between Natural Gas and Oil & NGL, (0.2301867 ,0.6960284) million metric tons.
3. Strong Evidence that state owned fuel use different than investor owned
4. Fugitive methane is a more reliable predictor than CO₂ when classifying between Natural Gas and Oil & NGL plants

Significance?

- Fugitive emissions should be a primary focus for distinguishing Natural Gas and Oil & NGL plants
- Carbon policy should prioritize methane over CO₂ for better classification and targeting
- Ownership structure should be considered when designing emissions regulations
- Investor vs state-owned differences suggest the need for specific energy-sector policies